

DMBI Unique Questions

Module I: Data Warehouse (DWH) Fundamentals with Introduction to Data Mining

#	Question	Marks	Month & Year
1	Explain types of attributes used in data exploration.	10	DEC 2023
2	Define data warehouse. Explain data warehouse architecture with help of a diagram	10	DEC 2023
3	Give the difference between OLAP and OLTP	10	DEC 2024
4	Compare star schema and snowflake schema	10	DEC 2024
5	Compare OLTP and OLAP operations	5	DEC 2025
6	Draw Data warehousing Architecture	3	MAY 2023
7	Compare and contrast between OLTP and OLAP	3	MAY 2023
8	Compare star schema, snowflake schema and star constellation	10	MAY 2023
9	With an example explain Star versus Snowflake schema in dimensional modelling	5	MAY 2024
10	Draw and list the components of a typical Data warehouse architecture	10	MAY 2024
11	What is OLAP? Explain various OLAP operations with neat labeled diagram	10	MAY 2024
12	Suppose that a data warehouse consists of dimensions time, doctor, and patient, and measures count and charge. Draw a star schema diagram. Starting with base cuboid [day, doctor, patient], what OLAP operations should be performed to list total fee collected by each doctor in 2004?	10	MAY 2025
13	Write short note on Dimensional Modeling	5	MAY 2023

Module II: Data Exploration and Data Preprocessing

#	Question	Marks	Month & Year
1	What is noisy data? How to handle it? For $D = \{ 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34 \}$, bins=3: i) equal frequency bins ii) smoothing by bin means iii) smoothing by bin boundaries	10	DEC 2023
2	What is noisy data? How to handle it? For $D = \{ 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34 \}$, bins=3: i) equal frequency bins ii) smoothing by bin means iii) smoothing by bin boundaries	10	DEC 2024
3	Different types of attributes with examples	5	DEC 2025
4	Explain different types of attributes used in data exploration with example	10	DEC 2024
5	What is noisy data? How to handle noisy data?	3	MAY 2023
6	Consider age of 29 participants: 5, 10, 13, 15, 16, 16, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70, 85. Explain & calculate mean, median, standard deviation, 1st & 3rd Quartile. Show Box and Whisker plot.	10	MAY 2023
7	A sales firm reported sales figures: 2300, 435, 675, 543, 454, 7877, 5434, 345, 2342, 654, 567, 545. Normalize using Min-Max scaling.	5	MAY 2024
8	Consider age of 35 participants (consider a sorted list). Draw histograms with bin size 5 and 8. Explain effect of bin size.	10	MAY 2024
9	Given data set [10, 20, 30, 40, 50]. Apply Min-Max normalization (range [0, 1]) and z-score normalization (for 30).	10	MAY 2025
10	Given scores of 10 students: [45, 50, 55, 58, 60, 65, 70, 72, 75, 80]. Create histogram with 3 bins. Show bin ranges and frequency counts graphically.	5	MAY 2025
11	Suppose data includes age values: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. i) Mean? ii) Median? iii) Mode & modality? iv) Mid-range? v) Five-point summary? vi) Box plot.	10	DEC 2025
12	Describe different steps involved in data preprocessing	10	DEC 2024
13	Write short note on types of attributes	5	MAY 2024

Module III: Classification

#	Question	Marks	Month & Year
1	Compare Bagging and Boosting of a classifier	10	DEC 2023
2	Using given training dataset (Homeowner, Marital Status, Job experience, Defaulted), classify tuple <Homeowner: No, Marital Status: Married, Job exp:3> using Naïve Bayes Algorithm	10	DEC 2023
3	Explain Confusion Matrix. Calculate Accuracy, Precision, Recall from given matrix (Yes/No classes)	10	DEC 2024
4	Using given training dataset (same as above) classify tuple using Naïve Bayes	10	DEC 2024
5	Explain bagging and boosting in classifiers in brief	10	DEC 2025
6	Explain concept of information gain and gini value used in decision tree algorithm	3	MAY 2023
7	Use Naïve Bayes to determine whether to play tennis on day with hot temp, rainy outlook, high humidity, no wind (given dataset)	10	MAY 2023
8	Explain working of decision tree based classifier. Explain steps for inducing tree using ID3 algorithm with example.	10	MAY 2024
9	Write short note on Random Forest Technique	5	MAY 2023
10	Write short note on Decision Tree Induction	5	MAY 2023
11	Write short note on Cross Validation	5	MAY 2023
12	Write short note on Naïve Bayes Classifier	5	MAY 2024
13	Write short note on Boot Strapping	5	MAY 2024
14	What is the difference between classification and prediction? Give examples.	5	MAY 2025
15	Given confusion matrix (Actual Yes/No vs Predicted Yes/No), calculate Accuracy and Precision	5	MAY 2025
16	Write a short note on simple linear regression	5	MAY 2025
17	Explain Simple Linear Regression with example	10	DEC 2025

#	Question	Marks	Month & Year
18	Explain Bagging of a classifier	10	DEC 2024

Module IV: Clustering and Outlier Detection

#	Question	Marks	Month & Year
1	Explain DBSCAN algorithm with example	10	DEC 2023
2	Explain K-means algorithm in detail. Apply to [2, 3, 6, 8, 9, 12, 15, 18, 22] into 3 clusters	10	DEC 2023
3	What is an outlier? List types of outliers. Describe methods used for outlier analysis	10	DEC 2023
4	Explain DBSCAN algorithm with example	10	DEC 2024
5	Explain K-means algorithm in detail. Apply to { 2, 3, 6, 8, 9, 12, 15, 18, 22 } into 3 clusters	10	DEC 2024
6	What is an outlier? List types. Describe methods for outlier analysis	10	DEC 2024
7	Explain DBSCAN algorithm using appropriate diagrams	10	DEC 2025
8	Using Hierarchical clustering algorithm to cluster 8 points A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9) into 3 clusters	10	DEC 2025
9	What is an outlier? Explain various methods for performing outlier analysis	10	MAY 2023
10	Cluster eight points A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9) into 3 clusters. Initial centers A1, A4, A7. Distance = Manhattan. Find cluster centers after second iteration (K-Means).	10	MAY 2023
11	What is an outlier? Explain various methods for outlier analysis	10	MAY 2024
12	Explain steps in hierarchical clustering. Perform hierarchical clustering on 10 points (2, 3), (5, 4), (9, 6), (4, 7), (8, 1), (7, 2), (6, 3), (1, 9), (3, 6), (4, 8) into 3 clusters.	10	MAY 2024
13	What is K-Means clustering?	5	MAY 2025

#	Question	Marks	Month & Year
14	Describe any two outlier detection methods	10	MAY 2025
15	Write short note on DBSCAN Algorithm	5	MAY 2023
16	Write short note on BIRCH Algorithm	5	MAY 2024

Module V: Frequent Pattern Mining

#	Question	Marks	Month & Year
1	Explain Multilevel and Multidimensional Association rules with suitable examples	10	DEC 2023
2	For the table given (TID: 011, 022, 031, 042, 051 with items), perform Apriori algorithm. Min support 30%, min confidence 70%. Show frequent itemsets and strong association rules.	10	DEC 2023
3	Explain Multilevel and Multidimensional Association rules with suitable examples	10	DEC 2024
4	For the table given (same as above), apply Apriori algorithm. Min support 30%, min confidence 70%.	10	DEC 2024
5	Explain Multidimensional and Multilevel association rule with example	10	DEC 2025
6	Explain Market-Basket analysis with example	10	DEC 2025
7	Explain market Basket Analysis with example	10	MAY 2023
8	Use Apriori algorithm to identify frequent itemsets from given database (Tid with items). Min support 50%, min confidence 75%. Extract strong association rules.	10	MAY 2023
9	Use Apriori algorithm to identify frequent itemsets from given database. Min support 75%, confidence 85%. Write all strong association rules.	10	MAY 2024
10	Explain mining of Multilevel association rules and Multidimensional association rules	10	MAY 2024
11	Consider transaction details (Milk, Bread, Diaper). Apply Apriori algorithm with min support 60%, confidence 75%. Find all frequent itemsets and association rules.	10	MAY 2025

#	Question	Marks	Month & Year
12	How to improve efficiency of Apriori algorithm?	5	MAY 2025
13	Describe Birch algorithm in detail	10	DEC 2025

Module VI: Business Intelligence

#	Question	Marks	Month & Year
1	Design BI system for Fraud Detection. Explain all steps from data collection to decision making	10	DEC 2023
2	Design BI system for Fraud Detection. Explain steps from data collection to decision making	10	DEC 2024
3	Define "Business Intelligence" and "Decision Support System" with example	5	DEC 2025
4	What is decision support system?	5	MAY 2025
5	Write short note on BI Architectures	5	MAY 2024
6	What are the key differences between Business Intelligence and traditional reporting systems?	5	MAY 2025